

Generation Interconnection System Impact Study Report

For Queue Project Q16-0698

Jacks Creek 345 kV

December 04, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0698. The Interconnection Customer proposes a 668.2 MW natural gas combined cycle facility interconnecting at the Jacks Creek 345 kV substation in the New Jersey area. The request was received on November 05, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 328 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,962,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gastonia Prime Power Park / Salton Sea 2 138 kV

Contingency: Union Gap / Jacks Crk 345 kV

Loading: 139.4% (Rating: 227 MVA)

DFAX: 11.7%

Pre-project loading: 104.9%

2. Facility: East 12Th Street / Salton Sea 2 138 kV line

Contingency: Union Gap / Jacks Creek 345 kV

Loading: 137.4% (Rating: 206 MVA)

DFAX: 11.7%

Pre-project loading: 99.5%

3. Facility: E 12Th Street / New Milford 138 kV line

Contingency: UnionGap / Jacks Crk 345 kV

Loading: 123.5% (Rating: 198 MVA)

DFAX: 11.7%

Pre-project loading: 84.0%

4. Facility: Closter / Lee Vining 161 kV

Contingency: Union Gap / Jacks Creek 345 kV

Loading: 110.8% (Rating: 290 MVA)

DFAX: 6.1%

Pre-project loading: 96.9%

5. Facility: Cuba City S / Deepcreek 161 kV

Contingency: Union Gap / Jacks Crk 345 kV

Loading: 110.0% (Rating: 278 MVA)

DFAX: 18.0%

Pre-project loading: 66.7%

6. Facility: Deepcreek / Eden 161 kV

Contingency: Union Gap / Jacks Crk 345 kV

Loading: 104.8% (Rating: 272 MVA)

DFAX: 18.0%

Pre-project loading: 60.6%

7. Facility: Lee Vining / Loch Miror 161 kV

Contingency: Union Gap / Jacks Crk 345 kV

Loading: 104.7% (Rating: 285 MVA)

DFAX: 6.1%

Pre-project loading: 90.6%

8. Facility: Wells Riv / Gastonia Prime Power Park 161 kV

Contingency: Union Gap Substation / Jacks Crk 345 kV line

Loading: 100.2% (Rating: 243 MVA)

DFAX: 12.9%

Pre-project loading: 64.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gastonia Prime Power Park / Salton Sea 2 138 kV	Rebuild line with higher-capacity conductor	\$644,000	100% (\$644,000)
East 12Th Street / Salton Sea 2 138 kV line	Rebuild line with higher-capacity conductor	\$1,136,000	100% (\$1,136,000)
E 12Th Street / New Milford 138 kV line	Rebuild line with higher-capacity conductor	\$2,516,000	100% (\$2,516,000)
Closter / Lee Vining 161 kV	Rebuild line with higher-capacity conductor	\$3,097,000	100% (\$3,097,000)
Cuba City S / Deepcreek 161 kV	Rebuild line with higher-capacity conductor	\$2,511,000	100% (\$2,511,000)
Eden / Deepcreek Substation 161 kV	Reconductor line	\$1,628,000	100% (\$1,628,000)
Lee Vining / Loch Miror 161 kV	Reconductor line	\$1,050,000	100% (\$1,050,000)
Gastonia Prime Power Park / Wells Riv 161 kV	Reconductor line	\$380,000	100% (\$380,000)
	Total allocated to project		\$12,962,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.